

Structured Module Lattice-based Cryptography

Fine-Grained Parameter Selection without Compromising Performance

Joppe W. Bos¹, Joost Renes², Frederik Vercauteren³ and Peng Wang⁴

¹ NXP Semiconductors, Leuven, Belgium

² NXP Semiconductors, Austin, USA

³ KU Leuven, Leuven, Belgium

⁴ University of Chinese Academy of Sciences, Beijing, China

Abstract. The ongoing transition to Post-Quantum Cryptography (PQC) has highlighted the need for cryptographic schemes that offer high security, strong performance, and fine-grained parameter selection. In lattice-based cryptography, particularly for the popular module variants of learning with errors (Module-LWE) and learning with rounding (Module-LWR) schemes based on power-of-two cyclotomics, existing constructions often force parameter choices that either overshoot or undershoot desired security levels due to structural constraints. In this work, we introduce a new class of techniques that are the best of both worlds: *structured* Module-LWE (or LWR) embeds *more* algebraic structure than a module such that it significantly improves performance, yet *less* structure than a power-of-two cyclotomic ring such that it still enables more flexible and efficient parameter selection. We present two concrete instances: a construction based on a radical extension of a two-power cyclotomic field denoted *radical* Ring-LWE (RR-LWE) or Ring-LWR (RR-LWR), and a cyclotomic block-ring module lattice approach (BRM-LWE or BRM-LWR). These new structured Module-LWE and LWR reduce the required number of uniformly random bytes in its matrix by a factor up to the module rank and allows efficient NTT implementations while enabling more granular security-performance trade-offs. We analyze the security of these constructions, provide practical parameter sets, and present implementation results demonstrating a performance improvement of up to 37% compared to an optimized implementation of ML-KEM. Our techniques apply to both key encapsulation mechanisms and digital signature schemes, offering a pathway to more adaptable and performant PQC standards.

Keywords: Post-quantum cryptography · Lattice-based cryptography · Parameter selection · Algebraic structure

1 Introduction

The cryptographic community is currently undergoing a transition from classical public-key cryptography to post-quantum cryptography (PQC). This shift is motivated by the anticipated emergence of large-scale quantum computers, which threaten to undermine the security of widely deployed schemes such as RSA [RSA78] and elliptic curve cryptography (ECC) [Kob87, Mil86]. As a result, significant efforts are underway to adapt cryptographic protocols and standards to incorporate quantum-resistant algorithms.

The United States National Institute of Standards and Technology (NIST) has recently finalized its first set of PQC standards, including the recommended module-lattice-based key encapsulation mechanism ML-KEM [MLK24] (based on CRYSTALS-Kyber [BDK⁺18]) and the recommended digital signature scheme ML-DSA [MLS24] (based on CRYSTALS-Dilithium [DKL⁺18]). However, the standardization process remains ongoing, as shown

Table 1: A non-exhaustive overview of Module-LWE and Module-LWR schemes of rank k working with polynomials of degree n . Columns n and k are highlighted to showcase what values are used in practice.

Name	Ref	LWE	LWR	n	k	NIST Level	Comment
Key Encapsulation Mechanisms							
SMAUG-T128	[HCCSS24], [CCHY24], [PJP+22]	✓	✓	256	2	1	Winner of Korean PQC effort
SMAUG-T192		✓	✓	256	3	3	Winner of Korean PQC effort
SMAUG-T256		✓	✓	256	4	5	Winner of Korean PQC effort
ML-KEM-512	[BDK+18], [MLK24]	✓		256	2	1	FIPS 203, CRYSTALS-Kyber
ML-KEM-768		✓		256	3	3	FIPS 203, CRYSTALS-Kyber
ML-KEM-1024		✓		256	4	5	FIPS 203, CRYSTALS-Kyber
NewHope512	[ADPS16]	✓		512	1	1	NIST Round-2
NewHope1024		✓		1024	1	5	NIST Round-2
Florete	[MKKV21]		✓	768	1	3	
Espada			✓	64	12	3	
Sable			✓	256	3	3	
LightSaber	[DKRV18]		✓	256	2	1	NIST Finalist
Saber			✓	256	3	3	NIST Finalist
FireSaber			✓	256	4	5	NIST Finalist
Digital Signature Algorithms							
ML-DSA-44	[DKL+18], [MLS24]	✓		256	4	2	FIPS 204, CRYSTALS-Dilithium
ML-DSA-65		✓		256	5	3	FIPS 204, CRYSTALS-Dilithium
ML-DSA-87		✓		256	7	5	FIPS 204, CRYSTALS-Dilithium
HAETA-120	[CCD+24]	✓		256	4	2	Winner of Korean PQC effort
HAETA-180		✓		256	6	3	Winner of Korean PQC effort
HAETA-260		✓		256	7	5	Winner of Korean PQC effort

by the “on-ramp” track for additional digital signature schemes, which aims to diversify the portfolio beyond lattice-based constructions [NIS25].

Beyond NIST, global and national PQC standardization initiatives are progressing in parallel. ISO/IEC is considering the inclusion of FrodoKEM [BCD⁺16, NAB⁺20] and Classic McEliece [ABC⁺22] alongside the NIST lattice-based KEMs in ISO/IEC 18033-2:2006 [Lon25]. In January 2025, South Korea concluded its KpqC competition, selecting four algorithms for future deployment [KN25]. China has announced plans to solicit proposals for PQC algorithms [Ins25], and India, among other countries, is expected to release its PQC guidelines and (recommendations for) standards in the near future.

Selecting appropriate parameters for cryptographic schemes is a well-known challenge [LV01], requiring a careful balance between security and practical performance. This trade-off is particularly pronounced in lattice-based cryptography, where security levels are closely tied to the selected parameters. For Ring-LWE [LPR10] and Ring-LWR [BPR12] a popular ring modulus of choice is the $2n$ -th cyclotomic polynomial $x^n + 1$, conveniently allowing the use of Number-Theoretic Transforms (NTT) [Pol71]. However, as this forces n to be a power of two, while the security is closely linked to n , it is difficult to accurately instantiate for a fixed security target.

To solve this problem, modern proposals rely instead on Module-LWE [Reg05, LPR10, BGV12, LS15] or Module-LWR [BPR12]. By adding a module rank k (which can be an arbitrary integer) on top of the cyclotomic ring structure, parameters can be selected more carefully. While reducing n and increasing k accordingly can yield a substantial security gain, it comes at the cost of a significant performance penalty. Since the matrix size is typically linear in n but quadratic in k , halving n and doubling k leads to twice the

matrix size. This impacts the matrix generation (often a performance bottleneck) as well as the matrix multiplication. Moreover, reducing n may require using multiple polynomial coefficient bits to encode the message, which increases the failure probability and ultimately lowers security. This limitation is well recognized. As noted in the ML-KEM specification:

“Smaller values of n would require to encode multiple key bits into one polynomial coefficient, which requires lower noise levels and therefore lowers security. Larger values of n would reduce the capability to easily scale security via parameter k .” [SAB⁺20]

Consequently, fine-grained selection of security parameters while maintaining strong performance remains a challenge, leading to either over- or under-provisioned security. A selection of popular Ring- and Module-LWE/LWR schemes with their choice for module dimension k and degree n is provided in Table 1. This highlights that often a degree of $n = 256$ is chosen which means much less flexibility to target a certain security level.

Contributions. We propose a solution for this problem with a new class of techniques, collectively referred to as *Structured Module-LWE* or *Structured Module-LWR*, that enable flexible and gradual parameter selection for Ring-LWE and Ring-LWR-based schemes while minimizing the impact on performance. Our approach achieves the same fine-grained parameter selection as Module-LWE and Module-LWR, yet removes the quadratic complexity in the matrix that leads to performance challenges. Replacing the module structure with one of our proposals for existing schemes such as ML-KEM, Saber, or ML-DSA, can lead to direct performance improvements. In short, we summarize our contributions as follows:

- We present *Radical* Ring-LWE (RR-LWE), which is an instance of Ring-LWE based on the polynomial $f(x) = (x^k - 2)^n + 1$, where n is a power of 2 and k an arbitrary positive integer. The number field defined by $f(x)$ is a degree k extension of the $2n$ -th cyclotomic field, thus admitting efficient arithmetic. We analyze its security, construct a concrete KEM from it and show that it exhibits a structure that is similar to Module-LWE, and that k plays a role similar to that of the module rank for fine-grained parameter selection, while offering better performance characteristics.
- We propose and analyze Block-Ring Module-LWE (BRM-LWE), which is a special case of Module-LWE consisting of matrices that retain a partial ring structure in the largest power-of-two submatrix. As it can be viewed as a special case of Module-LWE, the module rank can be used for fine-grained parameter selection, while the structure of the submatrix can be employed for efficient implementation.
- We include a reference and optimized implementation of a KEM based on RR-LWE and compare its performance to ML-KEM, demonstrating a performance improvement of 7-9% compared to ML-KEM-512, of 25-28% for ML-KEM-768, and of 31-37% for ML-KEM-1024 across `keypair`, `encaps` and `decaps` for modern Intel[®] Core[™] i7-9700K (Coffee Lake) CPUs including AVX2 support.

All proposals apply to LWE as well as LWR and can be used for constructions such as key encapsulation mechanisms and digital signature schemes where rings or module structure are currently used.

Organization. We start with preliminaries in Section 2, followed by a recollection of ring structures used for Ring and Module-LWE in Section 3 that forms the basis of our proposals. We present RR-LWE in Section 4 and BRM-LWE in Section 5. We provide implementation details and benchmarks in Section 6 followed by conclusions and considerations for future work in Section 7.

Source code. The source code for all implementations will be made available to enable reproducibility of the results.

2 Preliminaries

2.1 Notation

The ring of integers is denoted $\mathbf{Z}$ and for a positive integer q , $\mathbf{Z}_q$ denotes the quotient ring $\mathbf{Z}/q\mathbf{Z}$. Given a positive integer m , the m -th cyclotomic polynomial, denoted by $\Phi_m(x)$, is defined as the unique monic polynomial whose roots are the primitive m -th roots of unity, in particular,

$$\Phi_m(x) = \prod_{\substack{k=1 \\ \gcd(k,m)=1}}^m \left(x - e^{2\pi i \frac{k}{m}} \right),$$

where $e^{2\pi i \frac{k}{m}}$ for $\gcd(k, m) = 1$ are the primitive m -th roots of unity. It can be shown that $\Phi_m(x)$ has integral coefficients and from the definition it follows that $\deg(\Phi_m(x)) = \varphi(m)$, where $\varphi(m)$ is Euler's totient function, which counts the number of positive integers less than m that are relatively prime to m . For $m = 2n$ with n itself a power of 2, it is well known that $\Phi_m(x) = x^n + 1$. The polynomial quotient ring is denoted as $\mathbf{R}_q = \mathbf{Z}_q[x]/(f(x))$ where $f(x) \in \mathbf{Z}[x]$ is monic. The uniform distribution over a set S is denoted by $\mathcal{U}(S)$. The inner product between two vectors a and s is written as $\langle \mathbf{a}, \mathbf{s} \rangle$.

2.2 Learning with Errors / Rounding

The Learning with Errors (LWE) problem was introduced by Regev in [Reg05] and the more efficient and structured variants are known as Ring-LWE [LPR10] and Module-LWE [BGV12, LS15]. We refer the interested reader to the comprehensive surveys related to LWE [Reg10, Pei16a], and simply recall its definition in the module variant (which encompasses the plain and ring variants).

Let $\mathbf{K}$ be a number field of degree n and let $\mathbf{R} = \mathcal{O}_K$ be its ring of integers and denote $\mathbf{R}_q = \mathbf{R}/q\mathbf{R}$. Let χ denote a distribution over $\mathbf{R}$, and fix a module rank k . Let $\mathbf{s} \in \mathbf{R}_q^k$ be a secret vector and define the distribution $A_{\mathbf{s}, \chi}$ as

$$A_{\mathbf{s}, \chi} = (\mathbf{a}, \langle \mathbf{a}, \mathbf{s} \rangle + \mathbf{e} \bmod q) \in \mathbf{R}_q^k \times \mathbf{R}_q \quad (1)$$

where $\mathbf{a} \leftarrow \mathcal{U}(\mathbf{R}_q^k)$ is sampled uniformly at random and $\mathbf{e} \leftarrow \chi$ is the error term.

The *search LWE problem* is given access to polynomially many independent samples from $A_{\mathbf{s}, \chi}$, recover the secret $\mathbf{s}$, while the *decision LWE problem* is to distinguish between samples drawn from $A_{\mathbf{s}, \chi}$ and samples drawn uniformly from $\mathbf{R}_q^k \times \mathbf{R}_q$.

Note that setting $k = 1$ recovers the definition of the Ring-LWE problem over $\mathbf{R}_q$ and by choosing $\mathbf{K} = \mathbf{Q}$, the field of rationals, we recover the definition of Plain-LWE in dimension k . Other algebraically structured LWE variants have also been introduced such as Polynomial-LWE [SSTX09, RSW18], Order-LWE [BBPS19], and Middle-Product LWE [RSSS17]. A general framework of all these LWE variants over commutative base rings is given in [PP24].

The learning with rounding (LWR) problem was introduced in [BPR12] and can be seen as a derandomization technique for LWE. The distribution $A_{\mathbf{s}, \chi}$ defined in (1) is derandomized by replacing the addition of the error term $\mathbf{e}$, by a simple scale and round operation of the inner product $\langle \mathbf{a}, \mathbf{s} \rangle$. In particular, a modulus $p < q$ is chosen and the elements in $\mathbf{Z}_q$ are divided up into p contiguous intervals. The scale and round operation then maps an element of $\mathbf{Z}_q$ to its corresponding index in $\mathbf{Z}_p$. This rounded inner product

$\left\lfloor \frac{p}{q} \langle \mathbf{a}, \mathbf{s} \rangle \right\rfloor$ is denoted as

$$\lfloor \langle \mathbf{a}, \mathbf{s} \rangle \rfloor_p = \left\lfloor \frac{p}{q} \langle \mathbf{a}, \mathbf{s} \rangle \right\rfloor \in \mathbf{Z}_p.$$

Here $\mathbf{Z}_q$ (resp. $\mathbf{Z}_p$) is represented by $[0, q[$ (resp. $[0, p[$). Note that when $p \mid q$, the rounded version of the uniform distribution on $\mathbf{Z}_q$ is again uniform on $\mathbf{Z}_p$. The rounding operation can be extended to vectors / polynomials by applying it coefficient wise. Analogously to the ring and module variants of learning with errors, one can define ring and module versions of LWR.

3 Ring-LWE as a Structured Instance of Module-LWE

In this section $\mathbf{R}_{q,n}$ denotes $\mathbf{Z}_q[x]/(f(x))$ with $f(x)$ an irreducible polynomial of degree n . The typical choice in practice will be $f(x) = x^n + 1$ with n a power of 2 and $q = 1 \bmod 2n$, since this allows for fast arithmetic in $\mathbf{R}_{q,n}$ using FFT-based techniques.

In the setting of Module-LWE a single sample is of the form $\langle \mathbf{a}, \mathbf{s} \rangle + \mathbf{e}$, where $\mathbf{a}, \mathbf{s} \in \mathbf{R}_{q,n}^k$ and $\mathbf{e} \in \mathbf{R}_{q,n}$, with k the rank of the module. The samples are selected such that $\mathbf{a}$ is uniformly distributed and $\mathbf{s}$ and $\mathbf{e}$ are generated from a “small” distribution. If m samples are selected then the security level is expected to be equivalent to a Plain-LWE problem with matrix of size $mn \times kn$, assuming that the ring and module structure cannot be exploited. Compared to Plain-LWE [Reg05], both Module-LWE [BGV12, LS15] and Ring-LWE [LPR10] benefit from the additional ring and module structure to improve efficiency and reduce key and ciphertext sizes. By selecting $m = k$ one recovers the setting of ML-KEM and Saber (using LWR). For simplicity we assume that $m = k$ from here onwards.

The main advantage of Module-LWE is that k can be arbitrarily chosen. For example, ML-KEM and Saber select $k \in \{2, 3, 4\}$. For contrast, Ring-LWE based on a cyclotomic ring achieving the same security level would sample from $R_{q,kn}$ and therefore must restrict k to powers of two. On the other hand, the advantage of Ring-LWE is that a sample element $\mathbf{a}$ contains only kn different elements in $\mathbf{Z}_q$ (a single element $\mathbf{a} \in R_{q,kn}$) compared to k^2n for Module-LWE (k samples of elements in $\mathbf{R}_{q,n}^k$). The quadratic scaling of Module-LWE reduces the efficiency compared to Ring-LWE, but the greater flexibility with k allows for easier instantiations of concrete parameter sets.

It is well known how Ring-LWE corresponds to (or can be viewed as) a structured Module-LWE instance. For each power of two $k \mid n$ we have the isomorphism

$$\mathbf{R}_{q,n} = \mathbf{Z}_q[x]/(x^n + 1) \cong (\mathbf{Z}_q[y]/(y^{\frac{n}{k}} + 1)) [x]/(x^k - y) \quad (2)$$

by sending

$$s(x) = \sum_{i=0}^n s_i x^i \mapsto \sum_{j=0}^{k-1} s_j(y) \cdot x^j, \quad \text{where} \quad s_j(y) = \sum_{i=0}^{n/k-1} s_{ki+j} \cdot y^i.$$

This simply divides the polynomial s up into k parts $s_j(y)$, where each $s_j(y)$ contains the coefficients with degree equal to j modulo k .

For example, for $k = 2$ we have $s_0(y)$ containing the coefficients of *even* degree and $s_1(y)$ the coefficients of *odd* degree. Taking two elements $\mathbf{a}, \mathbf{s} \in \mathbf{R}_{q,n}$, under this isomorphism we see that

$$\mathbf{a} \cdot \mathbf{s} = \begin{bmatrix} a_0(y) & ya_1(y) \\ a_1(y) & a_0(y) \end{bmatrix} \begin{bmatrix} s_0(y) \\ s_1(y) \end{bmatrix},$$

where each operation takes place in the ring $\mathbf{Z}_q[y]/(y^{\frac{n}{2}} + 1)$, and in particular ya_1 corresponds to an anti-cyclic rotation of the coefficients of $a_1(y)$. This follows from the fact

that

$$\begin{aligned}
\mathbf{a} \cdot \mathbf{s} &= (a_0(y) + a_1(y)x) \cdot (s_0(y) + s_1(y)x) \\
&= a_0s_0(y) + (a_0s_1(y) + a_1s_0(y))x + a_1s_1(y)x^2 \\
&= a_0s_0(y) + ya_1s_1(y) + (a_0s_1(y) + a_1s_0(y))x \bmod (x^2 - y).
\end{aligned}$$

For $k = 4$ we would instead get

$$\mathbf{a} \cdot \mathbf{s} = \begin{bmatrix} a_0(y) & ya_3(y) & ya_2(y) & ya_1(y) \\ a_1(y) & a_0(y) & ya_3(y) & ya_2(y) \\ a_2(y) & a_1(y) & a_0(y) & ya_3(y) \\ a_3(y) & a_2(y) & a_1(y) & a_0(y) \end{bmatrix} \begin{bmatrix} s_0(y) \\ s_1(y) \\ s_2(y) \\ s_3(y) \end{bmatrix}, \quad (3)$$

operations now taking place in $\mathbf{Z}_q[y]/(y^{\frac{n}{4}} + 1)$. More generally, it is easy to see that $\mathbf{a} \cdot \mathbf{s}$ corresponds to a multiplication of a $k \times k$ circulant matrix with upper right triangle multiplied by y . For comparison, for $k = 4$, a Module-LWE matrix multiplication is structured as

$$\begin{bmatrix} a_{0,0}(y) & a_{0,1}(y) & a_{0,2}(y) & a_{0,3}(y) \\ a_{1,0}(y) & a_{1,1}(y) & a_{1,2}(y) & a_{1,3}(y) \\ a_{2,0}(y) & a_{2,1}(y) & a_{2,2}(y) & a_{2,3}(y) \\ a_{3,0}(y) & a_{3,1}(y) & a_{3,2}(y) & a_{3,3}(y) \end{bmatrix} \begin{bmatrix} s_0(y) \\ s_1(y) \\ s_2(y) \\ s_3(y) \end{bmatrix},$$

where arithmetic also takes place in the ring $\mathbf{Z}_q[y]/(y^{\frac{n}{4}} + 1)$.

To summarize, for any $k \mid n$ one can view a cyclotomic Ring-LWE sample as a structured Module-LWE sample of rank k where

$$a_{i,j}(y) = \begin{cases} \text{uniformly random,} & \text{for } j = 0, \\ y \cdot a_{k+i-j,0}(y), & \text{for } 0 \leq i < j < k, \\ a_{i-j,0}(y), & \text{for } 0 < j \leq i < k. \end{cases} \quad (4)$$

The above reinterpretation is purely syntactic and has no impact on the security. Under the ring isomorphism in Eq. (2), a cyclotomic Ring-LWE sample is mapped bijectively to the structured Module-LWE instance of rank k described in Eq. (4). Any distinguisher or secret-recovery algorithm for structured Module-LWE immediately yields one for Ring-LWE by composing with the inverse isomorphism, with identical advantage and essentially the same running time. Hence, viewing a cyclotomic Ring-LWE sample as a structured Module-LWE sample of rank k does not alter the underlying hardness assumptions nor concrete estimates: it is simply a different representation of the same distribution (and the same reasoning applies LWR).

This observation forms one of the key ingredients for us to define a different way to interpret and define new rings and parameters for Ring-LWE and Module-LWE based schemes.

4 Radical Ring-LWE

In this section we present the first proposal which contains the most structure but is also the most efficient. Let n be a power of two. Instead of a power-of-two cyclotomic $f(x) = x^n + 1$, we select the polynomial $f(x) = (x^k - 2)^n + 1$ for a strictly positive integer k .

In Lemma 1 below we show that f is irreducible, implying that $\mathbf{Q}[x]/(f(x))$ is a number field and as such we can define a Ring-LWE problem for this $f(x)$, the hardness of which is based on lattice problems in the corresponding ideal lattice in the canonical embedding [LPR10]. We refer to Section 4.1 for an in-depth discussion on how to instantiate Ring-LWE in this case.

Lemma 1. *Let $k \geq 1$ and $n = 2^m$ be integers for some $m > 0$, then*

$$f(x) = (x^k - 2)^n + 1 \in \mathbf{Z}[x]$$

is irreducible over $\mathbf{Q}$.

Proof. Set $y = x^k - 2$, then the irreducibility of $f(x)$ is equivalent to the fact that the polynomial $x^k - (\xi_{2n} + 2)$ is irreducible over the number field $\mathbf{K} = \mathbf{Q}[\xi_{2n}] = \mathbf{Q}[y]/(y^n + 1)$. According to the Vahlen-Capelli irreducibility criterion (proven in [Lan02, Ch. VI, Thm. 9.1]), a binomial of the form $x^k - a$ with $a \in \mathbf{K}$ for $\mathbf{K}$ a number field is irreducible if and only if for every prime $p \mid k$, we have that $a \notin \mathbf{K}^p$ and when $4 \mid k$, then $a \notin -4\mathbf{K}^4$.

Assume to the contrary that for $p \mid k$, there exists an element $\beta \in \mathbf{K}$ such that $\beta^p = \xi_{2n} + 2$. Taking field norms of both sides (which is multiplicative) shows that there exists an $m \in \mathbf{Z}$ such that $m^p = 2^n + 1$ (where the latter follows from the fact that the norm of $\xi_{2n} + 2 = 2^n + 1$, e.g. by taking resultants). Note that this implies that m is odd. When p is odd, we can write

$$2^n = m^p - 1 = (m - 1)(m^{p-1} + \dots + 1),$$

but the latter factor is a sum of p terms which are odd, and thus is also odd. Since this has to be a factor of 2^n it can only be 1, which is impossible since $m \geq 1$. For $p = 2$, we can write $m^2 - 1 = (m + 1)(m - 1) = 2^n$ and thus $m + 1 = 2^b$ and $m - 1 = 2^a$ with $b > a$ and $a + b = n$. Subtracting both gives $2 = 2^a(2^{b-a} - 1)$, which implies that $a = 1$ and $b = 2$, but this gives $n = 3$ which is not a power of 2.

Finally for the case $4 \mid k$, assume that $-4\beta^4 = 2 + \xi_{2n}$ for some $\beta \in \mathbf{K}$. Again taking field norms of both sides gives $2^{2n}m^4 = 2^n + 1$ where now $m \in \mathbf{Q}$ since β is not necessarily an algebraic integer. Since $n > 1$ is a power of 2, we have that $2n$ is divisible by 4 so we can rewrite $(2^{n/2}m)^4 = 2^n + 1$, where $2^{n/2}m \in \mathbf{Z}$. But we have already shown that $x^2 - 1 = 2^n$ has no solutions when n is a power of 2, so m does not exist. $\square$

The main reason for selecting f is due to the existence of the ring isomorphism

$$S_{q,n,k} := \mathbf{Z}_q[x]/((x^k - 2)^n + 1) \cong (\mathbf{Z}_q[y]/(y^n + 1)[x])/(x^k - y - 2), \quad (5)$$

$$x^k \mapsto y + 2.$$

While $S_{q,n,k}$ is not a cyclotomic ring itself, it has a large power-of-two cyclotomic subring in which arithmetic optimizations (such as NTTs) can still be used. The field defined by $f(x)$ is thus a degree k extension of the $2n$ -th cyclotomic field and the role of the extension degree k is similar to that of a module rank, allowing easier scaling of the hardness of the underlying lattice problem than for cyclotomic rings in which n must be a power of two. Since the polynomial $x^k - y - 2$ defines a degree k radical extension of the $2n$ -th cyclotomic field we refer to this type of ring as *radical* Ring-LWE (RR-LWE), or *radical* Ring-LWR (RR-LWR) in the learning with rounding setting.

Remark 1. Note that even if one would consider a more general m -th cyclotomic field instead of just a power of two n , it would still be impossible to construct cyclotomic fields with a large power-of-two cyclotomic subfield and of given degree. Indeed, since the degree of the m -th cyclotomic field is given by Euler $\varphi(m)$, the possible extension degrees are limited to integers of this form and even more so when a large power-of-two subfield is required for efficiency reasons. To illustrate this: since there are no integers m with $\varphi(m) = 3$ or $\varphi(m) = 5$, there do not exist cyclotomic fields that correspond to degree 3 or 5 extensions of a two-power cyclotomic field. With the above choice of $f(x)$ such fields are immediate simply by setting $k = 3$ or $k = 5$.

Compared to using a generic module structure, the above construction has the advantage of imposing a ring structure that reduces the computational complexity and randomness compared to Module-LWE. That is, the matrix $\mathbf{a}$ consists of k as opposed to k^2 uniformly random elements in the cyclotomic ring. One has to be careful how to instantiate Ring-LWE over a number ring different from a cyclotomic ring; in Section 4.1 we prove that our instantiation is in fact a secure instance of Ring-LWE.

More concretely, for elements $\mathbf{a}, \mathbf{s} \in S_{q,n,k}$ and writing $\mathbf{a} = \sum_{i=0}^{k-1} a_i(y)x^i$ and $\mathbf{s} = \sum_{i=0}^{k-1} s_i(y)x^i$, the polynomial multiplication $\mathbf{a} \cdot \mathbf{s}$ corresponds to the matrix multiplication

$$\begin{bmatrix} a_0(y) & (y+2) \cdot a_{k-1}(y) & (y+2) \cdot a_{k-2}(y) & \cdots & (y+2) \cdot a_1(y) \\ a_1(y) & a_0(y) & (y+2) \cdot a_{k-1}(y) & \cdots & (y+2) \cdot a_2(y) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{k-1}(y) & a_{k-2}(y) & a_{k-3}(y) & \cdots & a_0(y) \end{bmatrix} \begin{bmatrix} s_0(y) \\ s_1(y) \\ s_2(y) \\ \vdots \\ s_{k-1}(y) \end{bmatrix}. \quad (6)$$

All operations in this matrix multiplication take place in the cyclotomic ring $\mathbf{Z}_q[y]/(y^n + 1)$ and can use Number-Theoretic Transforms, while the multiplications with $(y+2)$ are a result of the reduction modulo $x^k - y - 2$. The multiplications with $y+2$ can be done with a single NTT to transform it to the NTT domain (or it can be pre-computed and stored), followed by $k-1$ base multiplications in the NTT domain.

Example 1 ($n = 256, k = 3$). Let us provide an example for the ring $S_{q,3,256}$ targeting a 192-bit security level comparable to ML-KEM-768 and Saber. In this case, we obtain a matrix multiplication

$$\begin{bmatrix} a_0(y) & (y+2) \cdot a_2(y) & (y+2) \cdot a_1(y) \\ a_1(y) & a_0(y) & (y+2) \cdot a_2(y) \\ a_2(y) & a_1(y) & a_0(y) \end{bmatrix} \begin{bmatrix} s_0(y) \\ s_1(y) \\ s_2(y) \end{bmatrix},$$

where all operations take place in $\mathbf{Z}_q[y]/(y^{256} + 1)$. The matrix requires about 1418 uniformly random bytes on average (using rejection sampling) to generate all a_i for the ML-KEM prime $q = 3329$, whereas ML-KEM-768 requires 4253 bytes on average: a reduction by a factor of $k = 3$. Using SHAKE128, this would need 14 additional Keccak permutations to squeeze the output. As the cost of Keccak is typically very significant (depending on the hardware), this is a meaningful improvement. This applies equally to the LWR setting using a power-of-two q . We refer to Table 4 for a comparison of the concrete run-times of ML-KEM and RR-LWE with an optimized implementation on a modern Intel processor using the AVX2 extension, showing that the cost of matrix generation is considerably reduced.

In the case of ML-KEM the matrix NTTs are implicitly computed as part of the sampling routine, so the arithmetic advantage is more limited. However, enforcing the NTT domain complicates the algorithm description and makes implementation techniques based on existing co-processors infeasible [BRV22]. The arithmetic benefit is more significant for LWR settings where sampling is done modulo a power-of-two q and, if NTTs in an extension are used for the multiplications, the NTT must be computed on each matrix element. Therefore a more significant improvement can be expected for performance compared to Module-LWR, especially as k grows.

PKE and KEM based on RR-LWE. As RR-LWE is a special instance of Ring-LWE, the PKE and KEM constructions, including proofs and reductions to Ring-LWE, remain unchanged. We summarize the full PKE in Algorithm 1, which can be transformed into a KEM using standard transformations. For simplicity, we do not include ciphertext compression in the description of the algorithm. The sampling of secrets can be done with distributions of

choice, for example a centered binomial to mimic the case of ML-KEM, and we assume $\mathbf{a}$ to be sampled uniformly at random. As usual, the notation $\mathbf{Sam}(\sigma)$ is used to highlight that sampling must be done deterministically based on the input seed σ .

Since we have two representations of the same ring $S_{q,n,k}$ given by (5), namely the first as the ring defined by $f(x)$ and the second as a degree k extension ring of the $2n$ -th cyclotomic ring, we have to specify what is meant by sampling from $\mathbf{Sam}(\sigma)$. As the notation in Algorithm 1 indicates, we are sampling k small elements in the ring $\mathbf{Z}_q[y]/(y^n + 1)$, which are then interpreted as coefficients of x^i for $i = 0, \dots, k-1$ defining an element in the second representation $(\mathbf{Z}_q[y]/(y^n + 1)[x])/(x^k - y - 2)$. As shown in [Pei16b], we note that sampling small coefficients in the first representation would not necessarily give rise to a secure instance of Ring-LWE due to the distortion when mapping from the canonical embedding (see Section 4.1 for more details).

Note that the established arithmetic shared secret $\mathbf{r} \cdot \mathbf{a} \cdot \mathbf{s}$ is an element of the full ring $S_{q,n,k} = \mathbf{Z}_q[x]/((x^k - 2)^n + 1)$. This allows encoding of (kn) -bit messages, which may be more than is required for a selected security level. Instead, the multiplications $\mathbf{r} \cdot (\mathbf{a} \cdot \mathbf{s})$ and $(\mathbf{r} \cdot \mathbf{a}) \cdot \mathbf{s}$ can be simplified to compute only a partial result such as only a subset of its coefficients. This reduces its complexity, and in case of a single coefficient would mimic the same computation as is done for Module-LWE. For example, setting $n = 128$ and $k = 4$ one targets a 128-bit security level and only one of the coefficients $\mathbf{r} \cdot \mathbf{a} \cdot \mathbf{s}$ is required to encode the 128-bit message. Writing $\mathbf{r}_i$ and $\mathbf{a}_i$ for the coefficients of $\mathbf{r}$ and $\mathbf{a}$ respectively, the most significant coefficient can be computed as $\sum_{i=0}^{k-1} \mathbf{r}_i \mathbf{a}_{k-1-i}$. This vector product consisting of k multiplications is simpler than the full polynomial multiplication $\mathbf{r} \cdot \mathbf{a}$, which is quadratic in k . In Algorithm 1 the additional option to select a subset of coefficients of the shared secret is incorporated through the use of the parameter ℓ . Additionally, the ciphertext size is reduced compared to regular Ring-LWE since $\mathbf{v}$ is only a partial ring element, requiring only ℓn coefficients to be transmitted instead of the full kn .

Again, a completely analogous RR-LWR construction can be designed based on Ring-LWR by using the same ring structure and rounding instead of adding an error vector.

4.1 Security

From the polynomial $f(x) = (x^k - 2)^n + 1$, we can define the standard Ring-LWE problem for the ring of integers $\mathcal{O}_{\mathbf{K}}$ of the number field $\mathbf{K} = \mathbf{Q}[x]/(f(x)) = \mathbf{Q}[\alpha]$, which is proven secure when instantiated properly, e.g. following for instance [Pei16b]. Therefore, the security assumption of RR-LWE is standard Ring-LWE, and we discuss the impact of the selection of a radical extension of a cyclotomic field below.

A secure Ring-LWE instance requires two components: first, the secret is assumed to be sampled from the dual ring $\mathcal{O}_{\mathbf{K}}^\vee$ and second, the noise is sampled from a (spherical) Gaussian distribution in the canonical embedding of $\mathbf{K}$. From $f(x)$ it is easy to explicitly write down the nk roots as follows $r_{i,j} = (\xi_{2n}^{(2j+1)} + 2)^{(1/k)} \xi_k^i$ for $i = 0, \dots, k-1$ and $j = 0, \dots, n-1$ and ξ_m a primitive m -th root of unity. Define an ordering of these roots such that $r_{m+nk/2} = \overline{r_m}$ for $m = 1, \dots, nk/2$, then the canonical embedding of an element $a(x) \in K$ consists of the nk evaluations $a(r_i)$ for $i = 1 \dots nk$.

Via an explicit computation (e.g. using Dedekind's criterion) one can verify that $\mathcal{O}_{\mathbf{K}} = \mathbf{Z}[x]/(f(x))$ is the ring of integers. Furthermore, the dual ring satisfies $\mathcal{O}_{\mathbf{K}}^\vee = \mathcal{O}_{\mathbf{K}}/(f'(\alpha))$ with $f'(\alpha)$ the derivative of f evaluated in α (recall $\mathbf{K} = \mathbf{Q}[\alpha]$). In this case we have $f'(x) = nkx^{(k-1)}(x^k - 2)$ and since $(\alpha^k - 2)$ is a unit in $\mathbf{K}$ (since $(\alpha^k - 2)^n + 1 = 0$) and multiplication by units does not change $\mathcal{O}_{\mathbf{K}}$, we conclude that $\mathcal{O}_{\mathbf{K}}^\vee = \mathcal{O}_{\mathbf{K}}/(nk\alpha^{(k-1)})$. A standard reduction then maps any Ring-LWE sample $(\mathbf{a}, \mathbf{a} \cdot \mathbf{s} + \mathbf{e} \bmod q)$ for a secret $\mathbf{s}$ from $\mathcal{O}_{\mathbf{K}}^\vee$ and $\mathbf{e}$ from χ (a spherical Gaussian on $\mathbf{K}$), to a sample $(\mathbf{a}, \mathbf{a}(ts) + t\mathbf{e})$ with $t = nk\alpha^{(k-1)}$, which is a Ring-LWE sample with secret $ts \in \mathcal{O}_{\mathbf{K}}$ and error distribution $t\chi$. The error distribution therefore is an elliptical error distribution where the i -th coordinate

Algorithm 1 Generic Public-Key Encryption (PKE) scheme based on RR-LWE.

RR-LWE-PKE.Keygen
Input: Parameters $q, n, k, D_s, D_e, D_{e_1}, D_{e_2}$
Output: (pk, sk)

- 1: $\rho, \sigma \leftarrow \{0, 1\}^{256}$
- 2: $\mathbf{a} \leftarrow S_{q,n,k} := \text{Sam}(\rho)$
- 3: $(\mathbf{s}, \mathbf{e}) \leftarrow D_s^k \times D_e^k := \text{Sam}(\sigma)$
- 4: $\mathbf{t} = \mathbf{a}\mathbf{s} + \mathbf{e}$
- 5: $pk := (\mathbf{t}, \rho), sk := \mathbf{s}$

RR-LWE-PKE.Encrypt
Input: $pk = (\mathbf{t}, \rho), \ell \in \{1, \dots, k\}, m \in \{0, 1\}^{n\ell}$
Output: c

- 1: $r \leftarrow \{0, 1\}^{256}$
- 2: $\mathbf{a} \leftarrow S_{q,n,k} := \text{Sam}(\rho)$
- 3: $(\mathbf{r}, \mathbf{e}_1, \mathbf{e}_2) \leftarrow D_s^k \times D_{e_1}^k \times D_{e_2}^\ell := \text{Sam}(r)$
- 4: $\mathbf{u} = \mathbf{r}\mathbf{a} + \mathbf{e}_1$
- 5: $\mathbf{v} = (\mathbf{r}\mathbf{t})_{k-\ell, \dots, k-1} + \mathbf{e}_2 + \lceil \frac{q}{2} \rceil \cdot m \quad \triangleright \ell \text{ most significant coefficients only}$
- 6: **return** $c := (\mathbf{u}, \mathbf{v})$

RR-LWE-PKE.Decrypt
Input: $(\mathbf{s}, c = (\mathbf{u}, \mathbf{v}))$
Output: $m' \in \{0, 1\}^{n\ell}$

- 1: **return** $m' := \lceil (2/q) \cdot (\mathbf{v} - \mathbf{u}\mathbf{s})_{k-\ell, \dots, k-1} \rceil \quad \triangleright \ell \text{ most significant coefficients only}$
-

is simply scaled by $\sigma_i(t) = nkr_i^{(k-1)}$ for $i = 1, \dots, nk$. Starting from a spherical Gaussian distribution χ on K , i.e. the probability distribution whose density is proportional to $\rho_\Sigma(x) = \exp(-\pi x^t \Sigma^{-1} x)$ with covariance matrix Σ a diagonal matrix with entries r^2 , the covariance matrix of the elliptical Gaussian $t\chi$ is thus given by $\Sigma' = \text{diag}(|\sigma_i(t)|^2 r^2)$. Since scaling by nk is the same for all coordinates, the only distortion is given by $|r_i^{(k-1)}|$. Using the expression for r_i and the fact that ξ_k is a unit, we have $|r_i^{(k-1)}| = |(\xi_{2n}^{(2j+1)} + 2)|^{(k-1)/k}$. Since $1 < |(\xi_{2n}^{(2j+1)} + 2)| < 3$ for all j , we conclude that the distortion is very small and limited to $3^{(k-1)/k} < 3$. More precisely, even for $k = 15$, the maximum distortion is less than 2.79.

Now that we have reduced to the case where the secret is sampled from $\mathcal{O}_K$, we still have to account for the difference between sampling coefficient-wise in the basis $y^i x^j$ for $i = 0, \dots, n-1$ and $j = 0, \dots, k-1$ in our ring representation $(\mathbf{Z}_q[y]/(y^n + 1)[x])/(x^k - y - 2)$ and the canonical embedding of $\mathbf{K}$. Note that this is a linear transformation L that can be obtained as the composition of two linear transformations:

- the first transformation maps the basis $y^i x^j$ into the corresponding element $b(x) = (x^k - 2)^i x^j \bmod f(x)$ (recall that indeed $y = x^k - 2$ in our ring)
- the second transformation is then the standard canonical embedding transformation mapping $b(x)$ into $[b(r_i)]_{i=1, \dots, nk}$. Note that this transformation is simply given by the Vandermonde matrix whose i -th row consists of the powers of r_i^j for $j = 0, \dots, nk-1$.

To bound the maximum distortion introduced by sampling in the basis $y^i x^j$ vs. in the canonical basis, it suffices to look at the minimum and maximum singular value of the above transformation L . Recall that in the "ideal case" of 2-power cyclotomic $x^n + 1$, the minimum and maximum singular value are the same, namely $\sqrt{n}$. We have computed the singular values of the transformation L for $k = 2, \dots, 15$ and for varying n and we find that the smallest singular value is $\sqrt{nk}$ (note that nk is the extension degree so this

is optimal) and the largest singular value is a small multiple $\beta\sqrt{nk}$ depending on k : for $k = 2$, $\beta < 1.74$, for $k = 3$, $\beta < 2.08$, for $k = 5$, $\beta < 2.41$, for $k = 7$, $\beta < 2.57$ and for $k = 15$, $\beta < 2.79$. This shows that sampling in the coefficient basis only incurs a minimal distortion vs. sampling in the canonical basis. As such we can conclude that the secret sampling and error distribution used result in a secure instance of Ring-LWE (for large enough dimensions). We can thus follow the usual approach (e.g. as is done for 2-power cyclotomics) and derive the security of that of generic lattice problems of dimension kn . For concrete instances we refer to Table 2 where we include example parameter sets for a KEM that match NIST security levels 1, 3 and 5. In particular, to highlight the advantage of our choice of ring, we include also parameter sets where kn is not a multiple of 256 (e.g., $kn = 640$ and $kn = 896$) which is not feasible for Ring-LWE with a power-of-two cyclotomic ring.

Finally, we note that since $f(x)$ is irreducible, the ring structure is not that of *multivariate* Ring-LWE named m-RLWE in [PUTPPG15] and a sequence of follow-up papers [PUTPPG16, APUPG17, PUTPG⁺21]. These works consider rings of the form $R_q[x, y] = \mathbf{Z}_q[x, y]/(g(x), h(y))$ where f and g are univariate polynomials defined over $\mathbf{Z}$. The m-RLWE problem was broken for certain choices of g, h in a paper from Bootland, Castryck and Vercauteren [BCV20], in particular when the roots of both g and h lie in a number field whose extension degree is smaller than $\deg(g)\deg(h)$. As the ring modulus $f(x) = (x^k - 2)^n + 1$ from Equation (5) really defines a number field of degree kn , these attacks do not apply. While we use the isomorphism from Equation (5) to make effective use of the ring structure, the lattice problem is simply a Ring-LWE problem over the *univariate* ring $S_{q,n,k}$.

4.2 Instantiations

Similar to Module-LWE, there is a lot of flexibility in the parameter selection for RR-LWE. The most obvious setting is that of Ring-LWE with a prime modulus, with a prime q such that $q \equiv 1 \pmod{2n}$ so that all n -th roots of unity exist in $\mathbf{Z}_q$. This allows NTTs to be used in the cyclotomic power-of-two ring $\mathbf{Z}_q[y]/(y^n + 1)$. This will be the focus of the parameter selection in this paper, as it allows to demonstrate the improvement compared to a state-of-the-art algorithm such as ML-KEM. We re-use the `lattice-estimator`¹, based on the estimator tool from Albrecht, Player and Scott [APS15], to estimate the security levels and failure probabilities for different choices of n and k . While n could be an arbitrarily small power of two, our minimal value in our parameter sets is $n = 64$ as it allows parameter selection at roughly 20-bit security intervals. All scripts and implementations work perfectly well for $n = 32$ and smaller. We refer to Table 2 for the full overview.

As we can see, we can select a parameter set with $n = 128$ and $k = 5$ which increases the classical and quantum core-SVP security compared to ML-KEM-512 by 28 and 26 bits, respectively. This helps push the security level comfortably over the requirements to meet NIST security level 1. Of course the same parameter sets could be defined for Module-LWE, but $k = 5$ times as much randomness would be required for the matrix generation compared to RR-LWE, making it much more costly. As $\mathbf{Z}_q$ contains all 256-th roots of unity, the same prime can be selected to compute NTTs in $\mathbf{Z}_q[y]/(y^n + 1)$. The secret key and error distribution are chosen to be centered binomials with similar values for η_1 and η_2 , and we assume ciphertext compression is applied to both $\mathbf{u}$ and $\mathbf{v}$. While the security increases for $k = 5$, this comes at the cost of an increased secret key (384 bytes), public key (192 bytes) and ciphertext (96 bytes) compared to ML-KEM-512. For completeness, we also include a parameter set targeting a 224-bit security level with $n = 128$ and $k = 7$. Alternatively, we could select $n = 64$ and $k = 9$, in which case $\ell = 2$ must be chosen to derive a 128-bit shared secret. As this allows for no security margin for

¹<https://github.com/pq-crystals/security-estimates>, commit 75c2694

Table 2: Parameter sets for Module-LWE, where ℓ is the RR-LWE and BRM-LWE parameter determining the size of the shared secret, δ is the failure probability, $\mathbf{C}$ and $\mathbf{Q}$ are the classical and quantum complexity of the Core-SVP instance respectively, $\mathbf{sk}$ is the secret key size, $\mathbf{pk}$ is the public key size, $\mathbf{ct}$ is the ciphertext size (all in bytes), and all other parameters use the same notation as the ML-KEM parameter sets [MLK24, Table 2].

	n	k	ℓ	q	η_1	η_2	(d_u, d_v)	δ	core-SVP		$\mathbf{sk}$	$\mathbf{pk}$	$\mathbf{ct}$
									$\mathbf{C}$	$\mathbf{Q}$			
ML-KEM	256	2	-	3329	3	2	(10, 4)	2^{-139}	117	106	1632	800	768
	256	3	-	3329	2	2	(10, 4)	2^{-165}	181	164	2400	1184	1088
	256	4	-	3329	2	2	(11, 5)	2^{-175}	253	230	3168	1568	1568
RR-LWE	256	2	1	3329	3	2	(10, 4)	2^{-139}	117	106	1632	800	768
	64	9	2	3329	2	2	(10, 4)	2^{-216}	128	116	1824	896	784
	128	5	1	3329	2	2	(10, 4)	2^{-196}	145	132	2016	992	864
	64	11	3	3329	2	2	(10, 4)	2^{-181}	163	148	2208	1088	976
	256	3	1	3329	2	2	(10, 4)	2^{-165}	181	164	2400	1184	1088
	64	13	4	3329	2	2	(10, 4)	2^{-155}	199	180	2592	1280	1168
	128	7	2	3329	2	2	(11, 5)	2^{-200}	217	197	2748	1376	1392
	64	15	4	3329	2	2	(11, 5)	2^{-188}	235	213	2976	1472	1480
	256	4	1	3329	2	2	(11, 5)	2^{-175}	253	230	3168	1568	1568

a 128-bit security level, $n = 128$ and $k = 5$ would be preferred.

As we shall see in Section 6, and in particular in Table 4, while the parameter sets can be chosen very close to those of ML-KEM, there are significant performance benefits in the selection of this ring. This advantage will be even greater for LWR and signature schemes where NTTs on the matrix are typically not pre-computed.

Field arithmetic. While the same prime $q = 3329$ is selected, and the same primitive root of unity $\zeta = 17$ can be used for NTTs, the NTT is not exactly equal for all parameter sets. As ML-KEM performs NTTs in the cyclotomic ring with modulus $x^{256} + 1$, while no 512-th root of unity is available in $\mathbf{Z}_q$, the NTT is only *partially* executed. That is, the polynomial is reduced only modulo $x^2 - \zeta^{2i+1}$ for $i \in \{1, \dots, 127\}$, corresponding to a 7-layer NTT. As a result, the base multiplication is actually a multiplication of quadratic polynomials, consisting of 5 multiplications in $\mathbf{Z}_q$. This means that while ML-KEM selects $n = 256$ and could theoretically benefit from a larger cyclotomic structure, it does not make full use of it due to the non-existence of 512-th roots of unity in $\mathbf{Z}_q$.

For RR-LWE we include parameter sets where $n = 128$, so NTTs take place in the ring $\mathbf{Z}_q[y]/(y^{128} + 1)$. We compute essentially the same 7 layers of butterflies (on half the coefficients), which now corresponds to a *full* NTT in the cyclotomic sub-ring. As a result, base multiplication can be performed coefficient-wise in $\mathbf{Z}_q$. We note that the same could have been done for ML-KEM by doing a single layer of schoolbook to decompose the 256-coefficient polynomial as 2 polynomials of 128 coefficients each, effectively skipping the first (as opposed to the last) layer of the full 8-layer NTT. Regardless, it shows that even though n is reduced from 256 to 128, the application of NTTs is essentially unchanged. For $n = 64$ and smaller only a subset of the NTT layers are executed, and fewer roots of unity are required in the base ring. One could select a prime smaller than q to reduce the computational effort, but since q is already relatively smaller for modern 32-bit or 64-bit systems, we have not pursued this direction.

Alternative design choices. Besides ML-KEM, other designs can similarly benefit from the RR-LWE structure. In essence RR-LWE serves as a drop-in replacement for Module-LWE, so has a wide variety of applications. As mentioned before, Ring-LWR can be instantiated based on $S_{q,n,k}$ in the same way. Taking the design from Saber [DKR⁺20], selecting a power-of-two q , and replacing Module-LWR by Ring-LWR based on $S_{q,n,k}$ would reduce the randomness required the (uniform) matrix sampling by a factor k . Similar to ML-KEM, the ciphertext size can be slightly reduced for LightSaber as only 128 bits of the message require encoding. Moreover, as Saber does not pre-compute matrix elements in the NTT domain, if a coefficient ring extension is used to apply NTTs to the polynomial multiplications, only k as opposed to k^2 NTTs are required to transform the matrix to NTT domain.

Since RR-LWE is independent from the secret (and error) distributions, other variants can also benefit. For example, one can use a sparse secret key distribution as done by SMAUG-T [HCCSS24] in the Korean PQC standardization effort. Finally, it can also be used for signature schemes such as ML-DSA [MLS24].

5 Block-Ring Module-LWE

In this section we propose the second variant and introduce Block-Ring Module-LWE (BRM-LWE) which aims to improve on Module-LWE by maintaining the flexibility of k , while removing the quadratic complexity. This approach is less efficient than RR-LWE introduced in Section 4 but it also intrinsically has less structure which could be a concern.

By observing that the Ring-LWE matrix is a special instance of Module-LWE (see Section 3), we create the BRM-LWE by restricting Module-LWE to the subset of samples that contain a partial ring structure. Assuming that samples from Ring-LWE and Module-LWE are indistinguishable, this leads to no loss of security. In particular, since no algorithms estimating the security of lattice-based proposals make specific use of the ring structure, there are no known attacks that affect BRM-LWE more than Module-LWE. We note that all arguments apply to both LWR as well as LWE.

5.1 BRM-LWE

As described in Section 3, a Ring-LWE structure over $\mathbf{R}_{q,kn}$ can be seen as a special case of Module-LWE with samples in $\mathbf{R}_{q,n}^k$ for powers of two k . In this section, we instead consider the case where k is *not* a power of two. We construct BRM-LWE by applying a Ring-LWE structure to the largest power-of-two submatrix. That is, writing $k = k_0 + k_1$ for integers k_0, k_1 where k_0 is a power of two (and k_1 does not have to be), we sample $\mathbf{s}$ from $\mathbf{R}_{q,n}^k$ as usual, but then consider the first k_0 entries as an element in $\mathbf{R}_{q,k_0n}$ using (2) and the last k_1 entries as an element of $\mathbf{R}_{q,n}^{k_1}$. We then construct a structured $\mathbf{a}$ as a $k \times k$ matrix with elements $a_{i,j} \in \mathbf{Z}_q[y]/(y^n + 1)$ with the additional condition that the $k_0 \times k_0$ block on the top left hand side follows the structure as defined in (4) and all other elements are uniformly random. By imposing this structure we can speed up the computation of the matrix vector multiply $\mathbf{a} \cdot \mathbf{s}$ (as well as $\mathbf{r} \cdot \mathbf{a}$ as shown later) by exploiting the fact that the multiplication of the $k_0 \times k_0$ block on the top left hand with the first k_0 entries of $\mathbf{s}$ corresponds precisely with multiplication in the ring $\mathbf{R}_{q,k_0n}$.

To illustrate the advantages of this construction, we provide a few concrete examples. Firstly, note that for $k_1 = 0$ (BRM-LWE) simply reduces back to Ring-LWE and for $k_0 = 0$ reduces to Module-LWE, showing that this is a *generalization* allowing for $k_0, k_1 \neq 0$.

$k = 4 + 1$. The first example revolves around the parameter selection for 128-bit security levels. For the combination $k = 2$ and $n = 256$, used by parameter sets in both ML-KEM and Saber, the security margin of Module-LWE over AES-128 is relatively small according

to the analysis from [SAB⁺22, Section 5.2]. This leaves some uncertainty in the security claims to reach NIST Security Level I, especially if lattice attacks were to improve over time. The obvious solution is to increase k , yet by the selection of $n = 256$ this leads to significantly overshooting the security (in fact, reaching NIST Security Level III for $k = 3$). Instead, in an ideal situation, one would like to simultaneously decrease n to $n = 128$ and select $k = 5$ for a more granular selection of the security level. However, this leads to the uniformly random matrix

$$\mathbf{a} = \begin{bmatrix} a_{0,0} & a_{0,1} & a_{0,2} & a_{0,3} & a_{0,4} \\ a_{1,0} & a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,0} & a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,0} & a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \\ a_{4,0} & a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{bmatrix}$$

consisting of $k^2 n = 3200$ random elements in $\mathbf{Z}_q$, compared to $4 \cdot 256 = 1024$ for $k = 2$ and $n = 256$. As random elements are typically generated using a XOF such as SHAKE128 which significantly contributes to the overall run-time, this impacts the efficiency of the algorithm. Additionally, more number-theoretic transformations (or possibly multiplications in case of LWR) are required as the top layer of the cyclotomic ring is removed. Moreover, since the shared secret in this setting is a polynomial with $n = 128$ coefficients, to achieve security levels greater than 128-bit one has to select multiple bits from the shared secret. This increases the failure probability, further complicating the parameter search and reducing the efficiency.

Instead, the proposed BRM-LWE constructs the matrix

$$\mathbf{a} = \begin{bmatrix} a_{0,0} & ya_{3,0} & ya_{2,0} & ya_{1,0} & a_{0,4} \\ a_{1,0} & a_{0,0} & ya_{3,0} & ya_{2,0} & a_{1,4} \\ a_{2,0} & a_{1,0} & a_{0,0} & ya_{3,0} & a_{2,4} \\ a_{3,0} & a_{2,0} & a_{1,0} & a_{0,0} & a_{3,4} \\ a_{4,0} & a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{bmatrix}$$

over the same ring, consisting of only $4 \cdot 128 + 9 \cdot 128 = 1664$ random elements in $\mathbf{Z}_q$ (a reduction by almost a factor of two). The secrets $\mathbf{s}$ and $\mathbf{r}$ are sampled as elements in $\mathbf{R}_{q,512} \times \mathbf{R}_{q,128}$, so that the multiplication of the square submatrix can be executed in the ring $\mathbf{R}_{q,512}$, improving the efficiency compared to a 4×4 matrix multiplication of quadratic complexity (for example, a 9-layer NTT could be used). Following the standard Module-LWE approach, the shared secret is an element in $\mathbf{R}_{q,n}$ which is computed as the inner product of $\mathbf{r}$ and $\mathbf{t} = \mathbf{a} \cdot \mathbf{s}$ in $\mathbf{R}_{q,n}^k$.

$k = 2 + 1$. To achieve a 192-bit security level, ML-KEM and Saber select $k = 3$ and $n = 256$. In this case, one can similarly use $k_0 = 2$ and $k_1 = 1$ to create a BRM-LWE construction that reduces the cost compared to Module-LWE. The structure of the matrix is then given by

$$\mathbf{a} = \begin{bmatrix} a_{0,0} & ya_{1,0} & a_{0,2} \\ a_{1,0} & a_{0,0} & a_{1,2} \\ a_{2,0} & a_{2,1} & a_{2,2} \end{bmatrix}$$

over $\mathbf{R}_{q,256}$, requiring only 1792 random elements in $\mathbf{Z}_q$ as opposed to 2304 for ML-KEM and Saber. This reduces the required number of random elements by almost a factor of two.

5.2 PKE and KEM based on BRM-LWE

In this section we complete the transformation from the BRM-LWE problem to a public-key encryption scheme and key encapsulation mechanism. The construction can be done

analogous to the design for ML-KEM, see [MLK24, Algorithm 13-15], and a standard variant of the Fujisaki-Okamoto transform [HHK17] can be applied to obtain an IND-CCA secure KEM. We highlight the differences when BRM-LWE is used.

In the setting of LWE, the public key becomes $\mathbf{t} = \mathbf{a} \cdot \mathbf{s} + \mathbf{e}$, where $\mathbf{t} \in \mathbf{R}_{q,n}^k$ and $\mathbf{e} \in \mathbf{R}_{q,k_0 n} \times \mathbf{R}_{q,n}^{k_1}$. Note that the distributions of the public key and error are not affected with respect to Module-LWE, but that part of the error is interpreted as a single element in $\mathbf{R}_{q,k_0 n}$ as opposed to k_0 elements of $\mathbf{R}_{q,n}$. This also holds for other errors that are sampled throughout the encryption process. To complete the encryption, one selects $\mathbf{r} \in \mathbf{R}_{q,k_0 n} \times \mathbf{R}_{q,n}^{k_1}$ and computes $\mathbf{u} = \mathbf{r} \cdot \mathbf{a} + \mathbf{e}_1 \in \mathbf{R}_{q,n}^k$, followed by the shared secret derivation as the inner product $\mathbf{v} = \langle \mathbf{r}, \mathbf{t} \rangle$. The decryption works by deriving and decoding $\mathbf{v} - \langle \mathbf{u}, \mathbf{s} \rangle$ as usual.

Note that the ring multiplication of the square submatrix in $\mathbf{r} \cdot \mathbf{a}$ corresponds to the polynomial multiplication where the coefficients of $\mathbf{r}$ are in reverse order, namely

$$\left(\sum_{i=0}^{k_0-1} \mathbf{r}_{k_0-1-i}(y)x^i \right) \cdot \left(\sum_{i=0}^{k_0-1} \mathbf{a}_{i,0}(y)x^i \right),$$

while the coefficients of the product are also in reverse order. This has no effect for the straightforward matrix multiplication

$$\begin{bmatrix} \mathbf{r}_0 & \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{r}_4 \end{bmatrix} \begin{bmatrix} a_{0,0} & ya_{3,0} & ya_{2,0} & ya_{1,0} & a_{0,4} \\ a_{1,0} & a_{0,0} & ya_{3,0} & ya_{2,0} & a_{1,4} \\ a_{2,0} & a_{1,0} & a_{0,0} & ya_{3,0} & a_{2,4} \\ a_{3,0} & a_{2,0} & a_{1,0} & a_{0,0} & a_{3,4} \\ a_{4,0} & a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{bmatrix},$$

but if a more efficient ring multiplication technique (such as an NTT) is used then this should be handled accordingly by interpreting the elements of $\mathbf{r}$ and $\mathbf{r} \cdot \mathbf{a}$ as polynomial coefficients as described above. Note that no explicit swaps are required as this can be solved by appropriately addressing the coefficients. We emphasize again that even if a standard matrix multiplication is used as would be done for Module-LWE, the reduction of entropy in the matrix itself still leads to a significant performance improvement.

For LWR the construction from Saber [DKR⁺20, Algorithm 1–6] can be re-used by rounding $\mathbf{a} \cdot \mathbf{s}$ instead of adding an error term. All matrix and vector multiplications are adapted in exactly the same manner.

Parameter selection follows the same process as RR-LWE, so we refer to Table 2 for an overview. A notable difference is that BRM-LWE is a special case of Module-LWE, in which the shared secret is only an element of $\mathbf{R}_{q,n}$, so the shared secret gets smaller as n decreases. Hence, to obtain a shared secret that is at least the length of the security parameter, multiple bits per coefficient may need to be selected. This is computationally straightforward, but will negatively affect the failure probability, so some care is required when generating parameters. As it is generally beneficial to select n as large as possible, we expect that this will have limited impact on practical instances.

5.3 Security

As BRM-LWE is a special instance of Module-LWE, in which the matrix has a specific structure, security reductions work analogously. We begin by stating the main security assumption for LWE. A similar security assumption can be stated in the setting of LWR.

Assumption 1 (Decision BRM-LWE). *Let $k = k_0 + k_1$ for a power of two k_0 and a positive*

integer k_1 . Let $\mathbf{a} = (\mathbf{a}_{i,j})_{0 \leq i,j < k}$ for $\mathbf{a}_{i,j} \in \mathbf{R}_{q,n} = \mathbf{Z}_q[y]/(y^n + 1)$ such that

$$\mathbf{a}_{i,j} = \begin{cases} \text{uniformly random,} & \text{for } j = 0, \\ y \cdot a_{k+i-j,0}(y), & \text{for } 0 \leq i < j < k_0, \\ a_{i-j,0}(y), & \text{for } 0 < j \leq i < k_0, \\ \text{uniformly random,} & \text{for } k_0 \leq i, j < k_0 + k_1. \end{cases}$$

For secret distribution D_s and error distribution D_e over $\mathbf{R}_{q,n}^k$, and for any algorithm A , the advantage of distinguishing $(\mathbf{a}, \mathbf{a} \cdot \mathbf{s} + \mathbf{e})$ and $(\mathbf{a}, \mathbf{b})$ for uniformly random $\mathbf{b} \in \mathbf{R}_{q,n}^k$ is negligible.

Note that the security assumption corresponds to Decision Ring-LWE over a power-of-two cyclotomic ring for $k_1 = 0$ and to Decision Module-LWE for $k_0 = 0$. Any algorithm solving BRM-LWE can trivially be used to solve Ring-LWE in $\mathbf{R}_{q,k_0n}$ by embedding the Ring-LWE sample into a BRM-LWE sample. Similarly, a Module-LWE instance can be embedded to solve Module-LWE in $\mathbf{R}_{q,n}^{k_1}$. In case $k_0, k_1 \neq 0$ we obtain a non-trivial BRM-LWE instance that has more ring structure than Module-LWE, yet less than a full ring, while retaining some of the computational advantages of Module-LWE. The parameter sets for RR-LWE from Table 2 can be re-used for BRM-LWE by additionally choosing values for k_0 and k_1 .

Finally, for $k_0 \neq 0$, Decision BRM-LWE is a special instance of Module-LWE with a block ring structure in the matrix. Hence, the question remains if this additional structure can be exploited. As the best known attacks on Module-LWE do not make use of the ring or module structure at all, the same attacks apply equally to BRM-LWE. We refer to [SAB⁺22, Section 5] for a more detailed discussion in the context of ML-KEM. Unless there are attacks that exploit the BRM-LWE structure, but do not affect Ring-LWE and Module-LWE, the security is equivalent. As we highlight the strong relation between BRM-LWE and the ring and module structures, we think it is unlikely that such attacks exist. Therefore we conjecture that BRM-LWE is as hard to solve as Plain-LWE with matrix size $kn \times kn$, as is the case for Ring-LWE and Module-LWE. Any proof demonstrating IND-CPA security of the PKE transfers without change. Similarly, any variant of a Fujisaki-Okamoto transformation to obtain an IND-CCA2 secure KEM equally applies.

6 Implementation

In this section we provide details of the implementation aspects of RR-LWE and compare benchmarks of optimized implementations against state-of-the-art ML-KEM software. We provide a pure C reference implementation that implements all parameter sets in Table 2, and an implementation optimized for the AVX2 instruction set targeting $n = 256$. Both implementations are based on the ML-KEM reference and optimized implementations², minimizing the changes to the selection of module or ring structure and therefore allowing a direct comparison under identical conditions. Our goal is to demonstrate that the advantage of RR-LWE is not limited to allowing more fine-grained parameter selection, but that for any of the security levels of ML-KEM a *performance gain* is achieved. In particular, for ML-KEM-1024, the run-time for all KEM operations is reduced by 31-37% for the optimized AVX2 implementation. We recall that the setup of ML-KEM is in fact a *worst case* comparison for RR-LWE, and even more significant performance improvements should be expected for Ring-LWR and digital signatures (like ML-DSA) where the matrix is not sampled in the NTT domain.

Setup. All benchmarks were performed on an Intel[®] Core™ i7-9700K (Coffee Lake) CPU running at 3.60GHz, with hyperthreading and TurboBoost disabled to ensure consistent

²<https://github.com/pq-crystals/kyber>, commit 4768bd3

timing results. All code was compiled from source (incl. ML-KEM) on Ubuntu 24.04.3 LTS with gcc v13.3.0 and compiler flag `-O3` to enable optimization for performance.

6.1 Reference Implementation

There are a few changes required for implementation of the parameter sets listed in Table 2. While $n = 256$ is supported in the ML-KEM reference implementation, smaller n are not. Luckily, the implementation makes use of the macro `KYBER_N` that allows relatively straightforward adaptation for most functions. However, the field arithmetic does not work out-of-the-box since it is specialized to a 7-layer NTT on a polynomial with 256 coefficients, with a 2-coefficient base multiplication. We include some minor changes to the `ntt`, `invntt` and `basemul` functions to support also polynomials of degree 128 and smaller, implementing the NTT as explained in Section 4.

For the ring structure the main modification is to update the `gen_matrix` function. As opposed to generating k^2 uniformly random elements, we generate only k elements $a_0, \dots, a_{k-1}$. The NTT of $y+2$ is pre-computed and stored and multiplied by $a_1, \dots, a_{k-1}$, though we leave a compile-time configuration option to compute it on-the-fly in case a smaller code size is preferred. To simplify integration we copy the elements into the $k \times k$ matrix as used for Module-LWE, according to Equation (6), so that most of the PKE and KEM operations can remain unchanged. While the copy operations could be avoided, we do not expect a significant impact on performance. Additionally, the inner product for `rt` and `us` is extended from a vector-vector multiplication to a polynomial multiplication that computes the ℓ most significant coefficients. This is included in the function `polyvec_partial_inner_prod`. In effect, this requires ℓk base multiplication as opposed to k multiplications for Module-LWE, increasing the cost if $\ell > 1$. As the cost is mainly determined by the NTT, and less so by the base multiplications, this impact is limited for optimized implementations. The parameter ℓ is denoted by `SS_NUM_COEFFS` in the implementation.

We provide an overview of the implemented parameters in our reference implementation in Table 3. It shows a gradual increase in security, with a similar increase in key and ciphertext sizes. We note that this implementation is not optimized for performance, but we include performance benchmarks for completeness. One can see an impact on the performance from the reduction of n , which is expected due to the quadratic cost of the pointwise multiplication since there is reduced cyclotomic structure. However, we note that this effect is exaggerated in the reference implementation as the cost of the arithmetic is relatively large compared to the SHAKE permutations. For example, the cost of generating a single byte using the `prf` function requires 1384 cycles for the reference implementation and 1042 (75%) cycles for the AVX2 optimized version. On the other hand, for $n = 256$ the NTT executes in 6851 cycles in the reference implementation and in only 153 cycles (2%) when optimized for AVX2. Moreover, the polynomial-wise generation of secrets and error terms loses some of its efficiency for small n , as more generated pseudo-random bytes are wasted. This can easily be improved by re-using the same Keccak state across multiple smaller polynomials. For a comparison of optimized implementations we refer to Table 4.

6.2 Optimized Implementation

Besides replacing the implementations of field arithmetic and SHAKE from C to vectorized AVX2 instructions, no further changes beyond what is described in Section 6.1 are technically required for the implementations. However, we observe that the SHAKE implementation uses `SHAKE128x4` to squeeze out the random bytes, which is a 4-way parallel implementation of SHAKE. This works well for $k = 2$ in the case of Module-LWE, as $k^2 = 4$ matrix elements are required. Each element is independently generated from the random input ρ and indices i, j that indicate the position in the matrix. In the case of

Table 3: Benchmarks for a *reference* implementation of the RR-LWE parameter sets defined in Table 2, where $N \times K$ refers to the parameters n and k , **C** and **Q** are the classical and quantum complexity of the Core-SVP instance respectively, **sk** is the secret key size, **pk** is the public key size, **ct** is the ciphertext size (all in bytes), and **keygen**, **encaps** and **decaps** are the measured runtimes in clock cycles averaged over 100 000 runs.

Parameter set	core-SVP		sk	pk	ct	keygen	encaps	decaps
	C	Q						
RR-LWE-256x2	117	106	1632	800	768	85 755	106 786	140 007
RR-LWE-64x9	128	116	1824	896	784	138 715	159 688	191 403
RR-LWE-128x5	145	132	2016	992	864	124 952	147 218	179 436
RR-LWE-64x11	163	148	2208	1088	976	188 055	227 158	274 966
RR-LWE-256x3	181	164	2400	1184	1088	137 245	157 486	200 352
RR-LWE-64x13	199	180	2592	1280	1168	223 050	272 334	333 808
RR-LWE-128x7	217	197	2748	1376	1392	185 678	223 741	281 821
RR-LWE-64x15	235	213	2976	1472	1480	294 326	345 471	421 541
RR-LWE-256x4	253	230	3168	1568	1568	192 292	229 276	287 718

Table 4: Average cycle count over 100 000 runs on a Intel® Core™ i7-9700K. ML-KEM figures are from the optimized AVX2 implementation and the RR-LWE approach is integrated in this optimized implementation.

	512		768		1024	
	ML-KEM	RR-LWE	ML-KEM	RR-LWE	ML-KEM	RR-LWE
gen_a	5 537	3 741	14 030	5 360	21 422	5 683
keypair	19 205	17 485	32 305	23 313	44 288	27 863
encaps	20 325	18 724	31 605	23 211	44 924	29 523
decaps	21 860	20 306	33 981	25 406	48 630	33 115

RR-LWE, only $k = 2$ elements are required so the use of **SHAKE128x4** in the same way is not as efficient. To improve the efficiency, we generate the upper and lower 128 coefficients of each polynomial independently for $k = 2$. While it reduces the efficiency of the sampling, as more unused bytes are sampled, the improvement is still worthwhile. If an efficient 2-way SHAKE implementation is available at double the throughput of **SHAKE128x4** that would be preferred, but we leave this optimization as an improvement for future work. For $k = 3$ and $k = 4$ we keep the matrix generation as-is.

6.3 Performance Comparison

Table 4 summarizes the average cycle counts over 100 000 runs for key operations in both ML-KEM and RR-LWE, across the three standardized parameter sets ($\{512, 768, 1024\}$). The operations measured include matrix generation (**gen_a**), keypair generation (**keypair**), encapsulation (**encaps**), and decapsulation (**decaps**). While the impact on the high-level functions **keypair**, **encaps** and **decaps** is most relevant for a practitioner, we include **gen_a** since it demonstrates the main source of the performance improvements.

Firstly, as shown, the RR-LWE approach consistently outperforms ML-KEM across all measured operations and parameter sets. We can see that the cost of matrix generation is reduced by 32%, 62% and 74% compared to ML-KEM-512, ML-KEM-768 and ML-KEM-1024 respectively. The absolute reduction in cycle counts for **gen_a** translated directly to an

improvement in `keypair`, `encaps` and `decaps`. As the relative cost of `gen_a` increases as k grows, the improvement of `RR-LWE` over `ML-KEM` also grows with k . While the advantage over `ML-KEM-512` is 7-9% for the KEM functions, it is 31-37% for `ML-KEM-1024`. This is a substantial reduction of the total run-time. Note that since `SHAKE128x4` is used, the difference between the matrix generation for $k = 3$ and $k = 4$ is relatively small.

Secondly, we also highlight the improved scalability of `RR-LWE`. For `ML-KEM`, increasing from the 512 to the 768 and 1024 parameter set incurs a performance impact of 55% and 122% for decapsulation respectively. As mentioned above, this is largely due to the increased matrix generation time that increases quadratically with k . In contrast, for `RR-LWE` the impact is only 25% and 63% respectively as the cost of `gen_a` increases only minimally (linearly with k , but the impact is skewed by the use of `SHAKE128x4`). This allows better scaling for larger parameter sets, and mitigates a lot of the performance impact of using smaller n that may be beneficial for hardware.

7 Conclusions and Future Work

We introduced new structured module lattice-based constructions `RR-LWE` and `BRM-LWE` that enable fine-grained parameter selection for both `LWE`- and `LWR`-based key encapsulation mechanisms and signature schemes. By embedding additional algebraic structure, our *structured module* `LWE/LWR` approaches allow practitioners to more precisely target desired security levels, overcoming the rigid granularity imposed by traditional schemes. Importantly, our implementation results demonstrate that these benefits are achieved without incurring a performance penalty: on the contrary, the proposed constructions exceed the efficiency of current standardized schemes.

The flexibility afforded by these techniques is particularly relevant in the context of ongoing and newly announced post-quantum cryptography standardization, where the ability to tailor parameters to specific security and performance requirements is increasingly valuable. Our results show that it is possible to achieve this adaptability while maintaining strong security assurances and practical efficiency.

Future work should focus on systematically exploring optimal parameter choices for both Plain/Ring/Module-`LWE/LWR`-based KEMs and signature schemes within this framework. This includes further analysis of security margins, failure probabilities, and implementation trade-offs across a broader range of deployment scenarios. Such investigations will help to fully understand the impact of this structured module approach in lattice-based cryptography and support the development of more versatile and robust post-quantum standards.

References

- [ABC⁺22] Martin R. Albrecht, Daniel J. Bernstein, Tung Chou, Carlos Cid, Jan Gilcher, Tanja Lange, Varun Maram, Ingo von Maurich, Rafael Misoczki, Ruben Niederhagen, Kenneth G. Paterson, Edoardo Persichetti, Christiane Peters, Peter Schwabe, Nicolas Sendrier, Jakub Szefer, Cen Jung Tjhai, Martin Tomlinson, and Wen Wang. Classic McEliece. Technical report, National Institute of Standards and Technology, 2022. available at <https://csrc.nist.gov/projects/post-quantum-cryptography/round-4-submissions>.
- [ADPS16] Erdem Alkim, Léo Ducas, Thomas Pöppelmann, and Peter Schwabe. Post-quantum key exchange - a new hope. In *USENIX security symposium*, pages 327–343, 2016.

- [APS15] Martin R. Albrecht, Rachel Player, and Sam Scott. On the concrete hardness of learning with errors. *Journal of Mathematical Cryptology*, 9(3):169–203, 2015. doi:[doi:10.1515/jmc-2015-0016](https://doi.org/10.1515/jmc-2015-0016).
- [APUPG17] Troncoso-Pastoriza Alberto Pedrouzo-Ulloa, Juan Ramón and Fernando Pérez-González. Multivariate cryptosystems for secure processing of multi-dimensional signals. *ArXiv*, abs/1712.00848, 2017.
- [BBPS19] Madalina Bolboceanu, Zvika Brakerski, Renen Perlman, and Devika Sharma. Order-LWE and the hardness of ring-LWE with entropic secrets. In Steven D. Galbraith and Shiho Moriai, editors, *ASIACRYPT 2019, Part II*, volume 11922 of *LNCS*, pages 91–120. Springer, Cham, December 2019. doi:[10.1007/978-3-030-34621-8_4](https://doi.org/10.1007/978-3-030-34621-8_4).
- [BCD⁺16] Joppe W. Bos, Craig Costello, Léo Ducas, Ilya Mironov, Michael Naehrig, Valeria Nikolaenko, Ananth Raghunathan, and Douglas Stebila. Frodo: Take off the ring! Practical, quantum-secure key exchange from LWE. In Edgar R. Weippl, Stefan Katzenbeisser, Christopher Kruegel, Andrew C. Myers, and Shai Halevi, editors, *ACM CCS 2016*, pages 1006–1018. ACM Press, October 2016. doi:[10.1145/2976749.2978425](https://doi.org/10.1145/2976749.2978425).
- [BCV20] Carl Bootland, Wouter Castryck, and Frederik Vercauteren. On the security of the multivariate ring learning with errors problem. In *ANTS-XIV*, volume 4, pages 57–71, 2020. doi:[10.2140/obs.2020.4.57](https://doi.org/10.2140/obs.2020.4.57).
- [BDK⁺18] Joppe W. Bos, Léo Ducas, Eike Kiltz, Tancrede Lepoint, Vadim Lyubashevsky, John M. Schanck, Peter Schwabe, Gregor Seiler, and Damien Stehlé. CRYSTALS - Kyber: A CCA-secure module-lattice-based KEM. In *2018 IEEE European Symposium on Security and Privacy*, pages 353–367. IEEE Computer Society Press, April 2018. doi:[10.1109/EuroSP.2018.00032](https://doi.org/10.1109/EuroSP.2018.00032).
- [BGV12] Zvika Brakerski, Craig Gentry, and Vinod Vaikuntanathan. (Leveled) fully homomorphic encryption without bootstrapping. In Shafi Goldwasser, editor, *ITCS 2012*, pages 309–325. ACM, January 2012. doi:[10.1145/2090236.2090262](https://doi.org/10.1145/2090236.2090262).
- [BPR12] Abhishek Banerjee, Chris Peikert, and Alon Rosen. Pseudorandom functions and lattices. In David Pointcheval and Thomas Johansson, editors, *EUROCRYPT 2012*, volume 7237 of *LNCS*, pages 719–737. Springer, Berlin, Heidelberg, April 2012. doi:[10.1007/978-3-642-29011-4_42](https://doi.org/10.1007/978-3-642-29011-4_42).
- [BRvV22] Joppe W. Bos, Joost Renes, and Christine van Vredendaal. Post-quantum cryptography with contemporary co-processors: Beyond Kronecker, Schönhage-Strassen & Nussbaumer. In Kevin R. B. Butler and Kurt Thomas, editors, *USENIX Security 2022*, pages 3683–3697. USENIX Association, August 2022. URL: <https://www.usenix.org/conference/usenixsecurity22/presentation/bos>.
- [CCD⁺24] Jung Hee Cheon, Hyeongmin Choe, Julien Devevey, Tim Güneysu, Dongyeon Hong, Markus Krausz, Georg Land, Marc Möller, Damien Stehlé, and MinJune Yi. HAETAE: Shorter lattice-based fiat-shamir signatures. *IACR TCHES*, 2024(3):25–75, 2024. doi:[10.46586/tches.v2024.i3.25-75](https://doi.org/10.46586/tches.v2024.i3.25-75).
- [CCHY24] Jung Hee Cheon, Hyeongmin Choe, Dongyeon Hong, and MinJune Yi. SMAUG: Pushing lattice-based key encapsulation mechanisms to the limits. In Claude Carlet, Kalikinkar Mandal, and Vincent Rijmen, editors, *SAC*

- 2023, volume 14201 of *LNCS*, pages 127–146. Springer, Cham, August 2024. doi:10.1007/978-3-031-53368-6_7.
- [DKL⁺18] Léo Ducas, Eike Kiltz, Tancrede Lepoint, Vadim Lyubashevsky, Peter Schwabe, Gregor Seiler, and Damien Stehlé. CRYSTALS-Dilithium: A lattice-based digital signature scheme. *IACR TCES*, 2018(1):238–268, 2018. URL: <https://tches.iacr.org/index.php/TCES/article/view/839>, doi:10.13154/tches.v2018.i1.238-268.
- [DKR⁺20] Jan-Pieter D’Anvers, Angshuman Karmakar, Sujoy Sinha Roy, Frederik Vercauteren, Jose Maria Bermudo Mera, Michiel Van Beirendonck, and Andrea Basso. SABER. Technical report, National Institute of Standards and Technology, 2020. available at <https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization/round-3-submissions>.
- [DKRV18] Jan-Pieter D’Anvers, Angshuman Karmakar, Sujoy Sinha Roy, and Frederik Vercauteren. Saber: Module-LWR based key exchange, CPA-secure encryption and CCA-secure KEM. In Antoine Joux, Abderrahmane Nitaj, and Tajjeeddine Rachidi, editors, *AFRICACRYPT 18*, volume 10831 of *LNCS*, pages 282–305. Springer, Cham, May 2018. doi:10.1007/978-3-319-89339-6_16.
- [HCCSS24] Jung Hee Cheon, Hyeongmin Choe, Jungjoo Seo, and Hyeon Seong. SMAUG(-T), revisited: Timing-secure, more compact, less failure. *IEEE Access*, 12:188386–188397, 2024. doi:10.1109/ACCESS.2024.3511346.
- [HHK17] Dennis Hofheinz, Kathrin Hövelmanns, and Eike Kiltz. A modular analysis of the Fujisaki-Okamoto transformation. In Yael Kalai and Leonid Reyzin, editors, *TCC 2017, Part I*, volume 10677 of *LNCS*, pages 341–371. Springer, Cham, November 2017. doi:10.1007/978-3-319-70500-2_12.
- [Ins25] Institute of Commercial Cryptography Standards. Announcement on launching the next-generation commercial cryptographic algorithms program (NGCC). <https://www.niccs.org.cn/en/>, 2025. Accessed: 2025-09-19.
- [KN25] Korean Information Security Agency and National Security Research Institute. Selected algorithms for the KpqC competition round 2. https://www.kpqc.or.kr/competition_02.html, 2025.
- [Kob87] N. Koblitz. Elliptic curve cryptosystems. *Mathematics of Computation*, 48:203–209, 1987.
- [Lan02] Serge Lang. *Algebra*, volume 211 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, revised 3rd edition, 2002.
- [Lon25] Patrick Longa. FrodoKEM: A conservative quantum-safe cryptographic algorithm. <https://www.microsoft.com/en-us/research/blog/frodokem-a-conservative-quantum-safe-cryptographic-algorithm/>, 2025. Accessed: 2025-09-19.
- [LPR10] Vadim Lyubashevsky, Chris Peikert, and Oded Regev. On ideal lattices and learning with errors over rings. In Henri Gilbert, editor, *EUROCRYPT 2010*, volume 6110 of *LNCS*, pages 1–23. Springer, Berlin, Heidelberg, May / June 2010. doi:10.1007/978-3-642-13190-5_1.

- [LS15] Adeline Langlois and Damien Stehlé. Worst-case to average-case reductions for module lattices. *DCC*, 75(3):565–599, 2015. doi:[10.1007/s10623-014-9938-4](https://doi.org/10.1007/s10623-014-9938-4).
- [LV01] Arjen K. Lenstra and Eric R. Verheul. Selecting cryptographic key sizes. *Journal of Cryptology*, 14(4):255–293, September 2001. doi:[10.1007/s00145-001-0009-4](https://doi.org/10.1007/s00145-001-0009-4).
- [Mil86] Victor S. Miller. Use of elliptic curves in cryptography. In Hugh C. Williams, editor, *CRYPTO’85*, volume 218 of *LNCS*, pages 417–426. Springer, Berlin, Heidelberg, August 1986. doi:[10.1007/3-540-39799-X_31](https://doi.org/10.1007/3-540-39799-X_31).
- [MKKV21] Jose Maria Bermudo Mera, Angshuman Karmakar, Suparna Kundu, and Ingrid Verbauwhede. Scabbard: a suite of efficient learning with rounding key-encapsulation mechanisms. *IACR TCHES*, 2021(4):474–509, 2021. URL: <https://tches.iacr.org/index.php/TCHES/article/view/9073>, doi:[10.46586/tches.v2021.i4.474-509](https://doi.org/10.46586/tches.v2021.i4.474-509).
- [MLK24] Module-lattice-based key-encapsulation mechanism standard. NIST FIPS PUB 203, U.S. Department of Commerce, 2024. doi:[10.6028/NIST.FIPS.203](https://doi.org/10.6028/NIST.FIPS.203).
- [MLS24] Module-lattice-based digital signature standard. NIST FIPS PUB 204, U.S. Department of Commerce, 2024. doi:[10.6028/NIST.FIPS.204](https://doi.org/10.6028/NIST.FIPS.204).
- [NAB⁺20] Michael Naehrig, Erdem Alkim, Joppe Bos, Léo Ducas, Karen Easterbrook, Brian LaMacchia, Patrick Longa, Ilya Mironov, Valeria Nikolaenko, Christopher Peikert, Ananth Raghunathan, and Douglas Stebila. FrodoKEM. Technical report, National Institute of Standards and Technology, 2020. available at <https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization/round-3-submissions>.
- [NIS25] NIST. Request for additional digital signature schemes for the post-quantum cryptography standardization process. <https://csrc.nist.gov/news/2022/request-additional-pqc-digital-signature-schemes>, 2025. Accessed: 2025-09-1.
- [Pei16a] Chris Peikert. A decade of lattice cryptography. *Foundations and Trends® in Theoretical Computer Science*, 10(4):283–424, 2016. doi:[http://dx.doi.org/10.1561/04000000074](https://dx.doi.org/10.1561/04000000074).
- [Pei16b] Chris Peikert. How (not) to instantiate ring-lwe. In Vassilis Zikas and Roberto De Prisco, editors, *Security and Cryptography for Networks*, pages 411–430, Cham, 2016. Springer International Publishing.
- [PJP⁺22] Seunghwan Park, Chi-Gon Jung, Aesun Park, Joongeun Choi, and Honggoo Kang. TiGER: Tiny bandwidth key encapsulation mechanism for easy migration based on RLWE(R). Cryptology ePrint Archive, Report 2022/1651, 2022. URL: <https://eprint.iacr.org/2022/1651>.
- [Pol71] John M. Pollard. The fast Fourier transform in a finite field. *Mathematics of Computation*, 25(114):365–374, 1971.
- [PP24] Chris Peikert and Zachary Pepin. Algebraically structured LWE, revisited. *Journal of Cryptology*, 37(3):28, July 2024. doi:[10.1007/s00145-024-09508-3](https://doi.org/10.1007/s00145-024-09508-3).

- [PUTPG⁺21] Alberto Pedrouzo-Ulloa, Juan Ramón Troncoso-Pastoriza, Nicolas Gama, Mariya Georgieva, and Fernando Pérez-González. Revisiting multivariate ring learning with errors and its applications on lattice-based cryptography. *Mathematics*, 9(8), 2021. URL: <https://www.mdpi.com/2227-7390/9/8/858>, doi:10.3390/math9080858.
- [PUTPPG15] Alberto Pedrouzo-Ulloa, Juan Ramón Troncoso-Pastoriza, and Fernando Pérez-González. Multivariate lattices for encrypted image processing. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1707–1711, 2015. doi:10.1109/ICASSP.2015.7178262.
- [PUTPPG16] Alberto Pedrouzo-Ulloa, Troncoso-Pastoriza, and Fernando Pérez-González. On ring learning with errors over the tensor product of number fields, 2016. URL: <https://arxiv.org/abs/1607.05244>.
- [Reg05] Oded Regev. On lattices, learning with errors, random linear codes, and cryptography. In Harold N. Gabow and Ronald Fagin, editors, *37th ACM STOC*, pages 84–93. ACM Press, May 2005. doi:10.1145/1060590.1060603.
- [Reg10] Oded Regev. The learning with errors problem (invited survey). In *IEEE Conference on Computational Complexity*, pages 191–204, 2010. doi:10.1109/CCC.2010.26.
- [RSA78] Ronald L. Rivest, Adi Shamir, and Leonard M. Adleman. A method for obtaining digital signatures and public-key cryptosystems. *Communications of the Association for Computing Machinery*, 21(2):120–126, February 1978. doi:10.1145/359340.359342.
- [RSSS17] Miruna Rosca, Amin Sakzad, Damien Stehlé, and Ron Steinfeld. Middle-product learning with errors. In Jonathan Katz and Hovav Shacham, editors, *CRYPTO 2017, Part III*, volume 10403 of *LNCS*, pages 283–297. Springer, Cham, August 2017. doi:10.1007/978-3-319-63697-9_10.
- [RSW18] Miruna Rosca, Damien Stehlé, and Alexandre Wallet. On the ring-LWE and polynomial-LWE problems. In Jesper Buus Nielsen and Vincent Rijmen, editors, *EUROCRYPT 2018, Part I*, volume 10820 of *LNCS*, pages 146–173. Springer, Cham, April / May 2018. doi:10.1007/978-3-319-78381-9_6.
- [SAB⁺20] Peter Schwabe, Roberto Avanzi, Joppe Bos, Léo Ducas, Eike Kiltz, Tancrede Lepoint, Vadim Lyubashevsky, John M. Schanck, Gregor Seiler, and Damien Stehlé. CRYSTALS-KYBER. Technical report, National Institute of Standards and Technology, 2020. available at <https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization/round-3-submissions>.
- [SAB⁺22] Peter Schwabe, Roberto Avanzi, Joppe Bos, Léo Ducas, Eike Kiltz, Tancrede Lepoint, Vadim Lyubashevsky, John M. Schanck, Gregor Seiler, Damien Stehlé, and Jintai Ding. CRYSTALS-KYBER. Technical report, National Institute of Standards and Technology, 2022. available at <https://csrc.nist.gov/Projects/post-quantum-cryptography/selected-algorithms-2022>.
- [SSTX09] Damien Stehlé, Ron Steinfeld, Keisuke Tanaka, and Keita Xagawa. Efficient public key encryption based on ideal lattices. In Mitsuru Matsui, editor, *ASIACRYPT 2009*, volume 5912 of *LNCS*, pages 617–635. Springer, Berlin, Heidelberg, December 2009. doi:10.1007/978-3-642-10366-7_36.